YEAR 10 MATHS · CHAPTER 5 · MIND MAP

Quadratic expressions and equations

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1 Expanding expressions

Multiply every term in one bracket by every term in the other.

$(x+3)(x-7) = x^2 - 4x - 21$

$(2x-5)^2 = 4x^2 - 20x + 25$

$(3x-1)(3x+1) = 9x^2 - 1$

2 Factorising expressions

Take out the HCF first, then look for the difference of two squares.

$12x^2 - 8x = 4x(3x-2)$

$25x^2 - 16 = (5x-4)(5x+4)$

$x^2 - 5 = (x-\sqrt{5})(x+\sqrt{5})$

3 Algebraic fractions: × and ÷

Factorise everything, cancel common factors, then multiply. To divide, flip the second fraction.

$\frac{x^2-9}{x+3} \times \frac{2}{x-3} = 2$

$\frac{x^2+2x}{4} \div \frac{x}{8} = 2(x+2)$

4 Monic trinomials

$x^2 + bx + c$: two numbers that multiply to c and add to b.

$x^2 - 2x - 15 = (x-5)(x+3)$

$x^2 + 9x + 20 = (x+4)(x+5)$

5 Non-monic trinomials

Find two numbers that multiply to ac and add to b, split the middle term, then group.

$3x^2 + 10x + 8$: $ac = 24$, use 6 and 4

$3x(x+2) + 4(x+2) = (3x+4)(x+2)$

6 Completing the square

Add and subtract $\left(\frac{b}{2}\right)^2$: $x^2 + bx = \left(x+\frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2$

$x^2 + 6x + 4 = (x+3)^2 - 5$

$= (x+3-\sqrt{5})(x+3+\sqrt{5})$

7 Solving by factorising

Make one side zero, factorise, then use the Null Factor Law.

$2x^2 - 5x - 3 = 0 \Rightarrow (2x+1)(x-3) = 0$

$x = -\frac{1}{2}$ or $x = 3$

8 Applications of quadratics

Write the equation, solve it, then check which answers make sense.

A ball's height is $h = 20t - 5t^2$. When is it 15 m up?

$5t^2 - 20t + 15 = 0 \Rightarrow t^2 - 4t + 3 = 0$, so $t = 1$ or 3 s

Two answers: once going up, once coming down.

9 Solving by completing the square

Complete the square, then take the square root of both sides (±).

$x^2 - 4x - 1 = 0 \Rightarrow (x-2)^2 = 5$

$x = 2 \pm \sqrt{5}$

10 The quadratic formula

$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$. The discriminant $\Delta = b^2 - 4ac$ gives the number of solutions.

$2x^2 + 3x - 1 = 0$: $\Delta = 9 + 8 = 17$, $x = \frac{-3 \pm \sqrt{17}}{4}$

$\Delta > 0$: two. $\Delta = 0$: one. $\Delta < 0$: none.